THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 16th one is: 1

(a) 32 **(b)** 272 **(c)** 240 **(d)** 256

2. 4 rays start from a point O, no two of them in the same direction. If every pair of rays makes one angle, how many angles are there? 1

(a) 3 **(b)** 6 **(c)** 12 **(d)** 4

3. The digit sum of 4567 is: 1

(a) 18 **(b)** 31 **(c)** 22 **(d)** 29

4. The number which occurs most often in the data 5, 2, 3, 2, 4, 2, 3, 2, 4, 1, 1, 3 is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 1

5. The prime factorisation of 72 has how many prime factors (counting repeats)? 1

(a) 2 **(b)** 6 **(c)** 5 **(d)** 4

6. Fencing a rectangular plot of perimeter 80 m costs ₹15 per metre. The total cost is (₹): 1

(a) 1185 **(b)** 1215 **(c)** 1300 **(d)** 1200

7. Arrange in ascending order: $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}$. 1

(a) $\frac{2}{3}, \frac{5}{6}, \frac{3}{4}$ **(b)** $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}$

(c) $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$ **(d)** $\frac{3}{4}, \frac{5}{6}, \frac{2}{3}$

8. A circle of radius 3 cm and a concentric circle of radius 7 cm are drawn. The width of the ring between them (cm): 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 4

9. A regular polygon has 7 lines of symmetry. The number of its sides is: 1

(a) 7 **(b)** 9 **(c)** 8 **(d)** 6

10. A diver at -15 m (below sea level) swims up 6 m. His new position (m) is: 1

(a) 3 **(b)** -9 **(c)** -7 **(d)** -11

11. In the pattern $1, 3, 6, 10, \ldots$ what is the 9th term? 1

(a) 81 **(b)** 36 **(c)** 55 **(d)** 45

12. A protractor shows the arm of an angle at 35° on the inner scale. The reading on the outer scale for the same arm is: 1

(a) 325 **(b)** 145 **(c)** 35 **(d)** 55

13. In a supercell pyramid each number is the sum of the two numbers just below it. If the two numbers below are 13 and 18, the number above them is: 1

(a) 234 **(b)** 41 **(c)** 31 **(d)** 5

14. In the data 1, 5, 2, 2, 4, 4, 4, 4, 4, 3, 2, 2, how many values are greater than 3? 1

(a) 5 **(b)** 6 **(c)** 8 **(d)** 7

15. Using the Sieve of Eratosthenes, which is the first prime whose multiples we cross out after crossing multiples of 2? 1

(a) 1 **(b)** 4 **(c)** 3 **(d)** 5

16. How many square tiles of side 20 cm are needed to cover a floor of 400 cm by 300 cm? 1

(a) 35 **(b)** 6000 **(c)** 300 **(d)** 70

17. A pizza is cut into 12 equal slices. Riya eats 4 slices. The fraction she eats in simplest form has denominator: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 5

18. The diagonals of a square are: 1

(a) unequal
(b) equal but not perpendicular
(c) parallel
(d) equal and perpendicular bisectors of each other

19. Which of these letters has rotational symmetry of order 2 but NO line of symmetry? 1

(a) H **(b)** A **(c)** O **(d)** Z

20. The sum of an integer and its opposite is always: 1

(a) 1 **(b)** 0
(c) the integer itself **(d)** twice the integer

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Do one Kaprekar step on 2961. 2

22. A bar graph is drawn with a scale of 1 cm = 5 students for the data below. Find the height of each bar. 2

Item	Number
Football	35
Chess	15
Hockey	20

23. Write two prime numbers whose sum is 24. 2

24. A rectangle of length 12 cm and breadth 5 cm is cut along a diagonal. Find the area of each triangle. 2

25. Subtract $\frac{1}{3}$ from $\frac{5}{6}$. 2

26. In rectangle ABCD the diagonals meet at O. If AC = 12 cm, find OA, OB, BD. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. 3 metres of cloth is cut into pieces of $\frac{2}{5}$ m each. How many pieces are made and how much is left? 3

28. In a square with diagonal intersection O and side 4 cm, state the angle between the diagonals and the angle $\angle OAB$. 3

29. For the capital letter A, state whether it has 3
(i) a vertical line of symmetry,
(ii) a horizontal line of symmetry,
(iii) rotational symmetry of order 2.

30. The temperature at 6 am is -4°C. It rises by 9°C by noon and then falls by 8°C by evening. Find the evening temperature. Is it below 0°C? 3

31. Look at the sequence $5, 7, 9, 11, 13$. Find the pattern and the 10th term. 3

OR

How many right angles make 180 degrees? What angle do 3 right angles make together? 3

32. A number is formed by writing the digit 7 exactly 5 times in a row. Find its digit sum and say whether the number is divisible by 3. 3

OR

Which scale is better to draw a bar graph for values 20, 40, 60: 1 cm = 1 unit or 1 cm = 10 units? Give the height of the largest bar with your chosen scale. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Letters of the word MATH: find which letters have vertical symmetry, which have horizontal symmetry and which have rotational symmetry of order 2. Is the word itself symmetric about a vertical mirror? 5

34. A shopkeeper records profit as positive and loss as negative: Mon 120, Tue -80, Wed 50, Thu -30 (in ₹). Find the total, the best day and the worst day. 5

35. A group of 7 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

36. A ray OA makes 45° with OB and another ray OC makes 135° with OB on the same side of OB, with OC beyond OA. Find $\angle AOC$, and name the types of $\angle AOB$, $\angle BOC$ and $\angle AOC$. 5

OR

Take the digits of 2718. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

37. The table shows data on favourite sports of students. Draw a pictograph taking one symbol = 5 students, and find the total number of symbols. Data: Cricket 20, Football 25, Hockey 15. 5

OR

Find all prime numbers between 41 and 80 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

38. An L-shaped figure is made by joining a rectangle 12 cm by 5 cm and a rectangle 6 cm by 4 cm (they do not overlap). Find the total area of the figure and by how much the larger part exceeds the smaller part. **5**

OR

Riya reads $\frac{2}{5}$ of a book on Monday and $\frac{1}{4}$ of it on Tuesday. What fraction has she read in two days? What fraction is left? On which day did she read more? **5**

END OF QUESTION PAPER **Total: 80 Marks**

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